



Detecting Credit Card Fraud Using Statistical Distance Measures

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Motivation

Credit card fraud costs the global economy billions of dollars annually, representing a significant and growing threat to financial systems worldwide. Traditional detection systems rely on fixed rules that flag transactions exceeding predefined thresholds for specific features. The fundamental problem is that these approaches cannot learn evolving patterns of normal behavior, making them ineffective against sophisticated and adaptive fraud strategies.



Source: Joakim Arvidsson

This project investigates whether modeling normal transaction behavior as a statistical distribution and measuring deviations using Wasserstein distance can provide a more adaptive fraud detection framework.

Data and Methods

The dataset is made by the Machine Learning Group at Université Libre de Bruxelles (ULB), released by Dal Pozzolo et al. (2014)¹. It contains transactions made by European cardholders over two days in September 2013.

- 284,807 total transactions, 492 confirmed fraud cases (0.172%)
- 30 features: 28 PCA-transformed components (V1–V28), transaction Amount, and Time

*Features anonymized via PCA due to confidentiality constraints

Time	V1	V2	V3
0.0	-1.359807	-0.072781	2.536347
0.0	1.191857	0.266151	0.166480

V21	V22	V23	V24
-0.018307	0.277838	-0.110474	0.066928
-0.225775	-0.638672	0.101288	0.339846

V27	V28	Amount	Class
0.133558	-0.021053	149.62	0
-0.014724	0.014724	2.69	0

Statistical Learning Algorithm

The Wasserstein distance measures the minimum cost of transforming one probability distribution into another. For fraud detection, it quantifies how far a new transaction deviates from learned normal behavior.

1. Feature Selection Two-sample t-tests between normal and fraud transactions, corrected with Benjamini-Hochberg FDR. 27 out of 30 features selected.

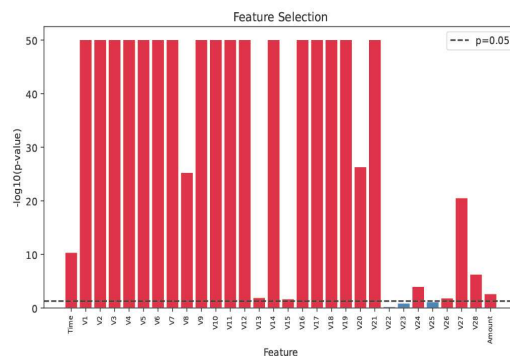
2. Distribution Fitting A Gaussian distribution fitted to each selected feature of normal transactions using PyTorch's L-BFGS optimizer.

3. Anomaly Scoring Wasserstein distance computed analytically per feature, weighted by discriminative power, and summed into a single anomaly score:

$$W_1(\delta_x, \mathcal{N}(\mu, \sigma)) = \sigma|z| + 2\sigma\phi(|z|)$$

where $z = \frac{x-\mu}{\sigma}$ and ϕ is the standard Gaussian PDF. Weighted distances are summed into a single anomaly score.

4. Threshold Optimization 300 candidate thresholds swept across the test set. Threshold maximizing F1-score selected as the decision boundary.



Limitations

- Severe class imbalance (578:1) cause fraud and normal score distributions to overlap significantly.
- Independent Gaussian assumption per feature ignores correlations between features.
- Method detects deviation from normal behavior but cannot learn fraud-specific patterns without labeled.

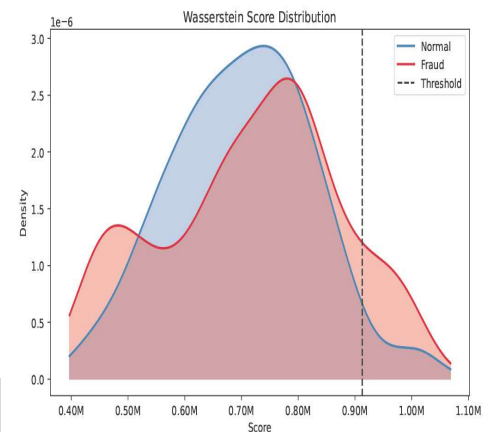
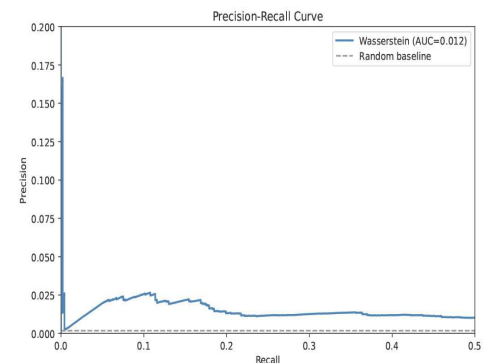
Future Work

- Fit multivariate distributions to capture inner-feature correlations.
- Compare other statistical distance such as KL divergence and Total Variation distance.

Results

- The Wasserstein anomaly score successfully assigns higher scores to fraud transactions on average, confirming that the method captures distributional deviation.
- However, the severe class imbalance limits binary classification performance

Method	Precision	Recall	F1 Score
Naive	0.0	0.0	0.0
Wasserstein	0.025629	0.107724	0.041406



Confusion Matrix		
	Pred Normal	Pred Fraud
Normal	54848 (96.5%)	2015 (3.5%)
Fraud	439 (89.2%)	53 (10.8%)

References

Dal Pozzolo, A., et al. (2014). Learned lessons in credit card fraud detection. *Expert Systems with Applications*, 41(10).